CHAPTER 4

THE FOURIER INTEGRAL

In the preceeding chapter we saw how a periodic signal can be represented as a linear combination of harmonically related exponentials. In fact, this idea can be extended to represent aperiodic signals in a similar manner.

4.1. THE FOURIER TRANSFORM REPRESENTATION OF AN APERIODIC SIGNAL

Let us start our development by an example and consider the periodic square wave shown in Figure 4.1.1. The Fourier series coefficients of this periodic signal are given as

$$a_k = \frac{2\,sin\,k\omega_0 t}{k\omega_0 t} \tag{4.1.1}$$

where T_0 is the period and $\omega_0 = 2\pi/T_0$.

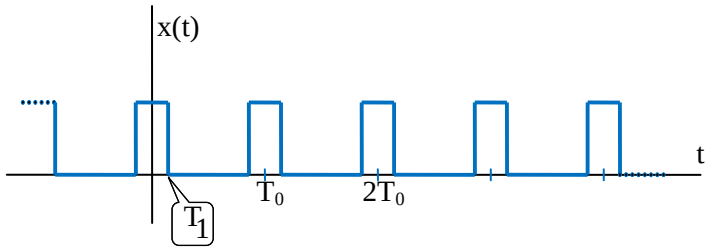

Fig. 4.1.1. Periodic square wave

Figure 4.1.2 depicts a plot of $T_0 a_k$ versus ω for a fixed value of T_1 and three different values of T_0. What is of interest here, is that the envelope, shown by dotted lines is independent of the period. As T_0 is increased, the envelope is sampled at a faster rate. As T_0 becomes arbitrarily large, the periodic square wave approaches a single pulse; an aperiodic signal that is equal to a single period of the square wave. On the other hand, the spectrum multiplied by T_0 , becomes more and more closely spaced samples of the envelope and at the limit becomes the envelope itself.

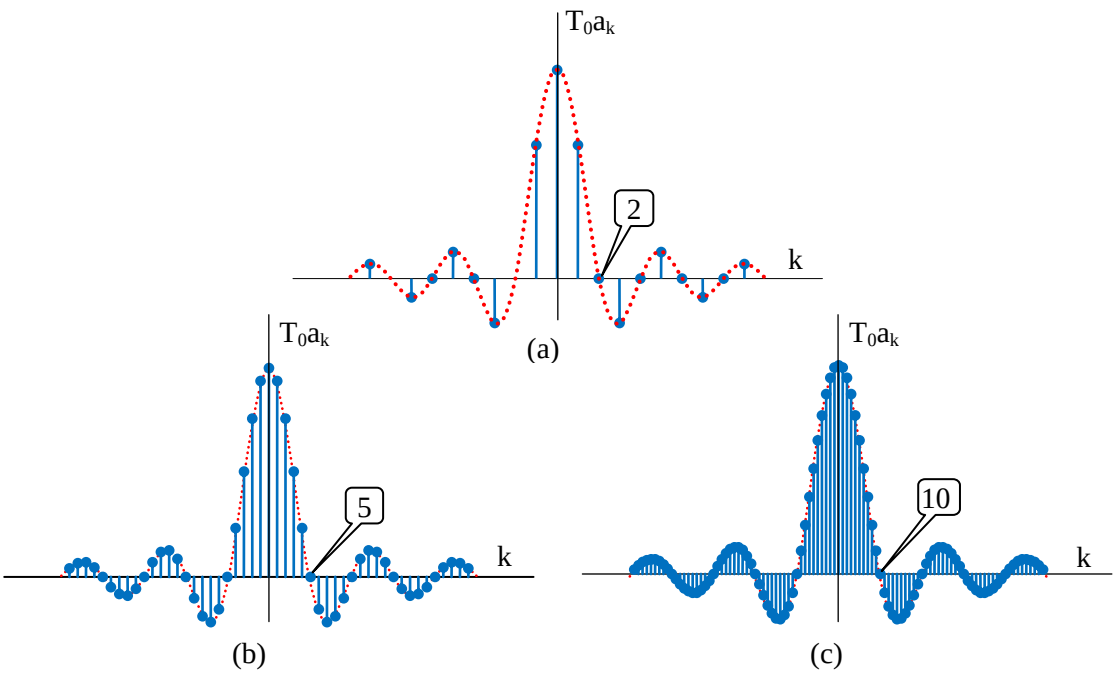

Fig. 4.1.2. Fourier coefficients for periodic square wave: (a) $\mathbf{T_0=4T_1}$, (b) $\mathbf{T_0=10T_1}$, (c) $\mathbf{T_0=20T_1}$

The basic idea behind the development of the Fourier transform is as explained by the example above: an aperiodic signal can be thought of as the limit of a periodic signal as the period becomes arbitrarily large. Let us consider an aperiodic signal of finite duration as shown in Figure 4.1.3(a). We can construct a periodic signal $\tilde{x}(t)$ from this aperiodic signal by repeating it every T_0 seconds, as shown in Figure 4.1.3(b). As we choose T_0 larger, the "pulses" in $\tilde{x}(t)$ become more widely separated and as T_0 approaches infinity, $\tilde{x}(t)$ approaches $x(t)$.

The periodic signal $\tilde{x}(t)$ has the Fourier series representation

$$\tilde{x}(t) = \sum_{k=-\infty}^{\infty} a_k \, e^{jk\omega_0 t} \tag{4.1.2a}$$

with

$$a_k = \frac{1}{T_0}\int_{-T_0/2}^{T_0/2} \tilde{x}(t) e^{jk\omega_0 t}\, dt \,. \tag{4.1.2b}$$

Since $\tilde{x}(t) = x(t)$ within the interval of integration and $x(t) = 0$ outside this interval, (4.1.2b) can be rewritten as

$$T_0 a_k = \int_{-\infty}^{\infty} x(t) e^{jk\omega_0 t} dt \quad . \tag{4.1.3}$$

Clearly (4.1.3) defines the samples of

$$X(\omega) = \int_{-\infty}^{\infty} x(t) e^{-j\omega t} dt \tag{4.1.4}$$

taken at $\omega = k\omega_0$, i.e.

$$T_0 a_k = X(k\omega_0) \ . \tag{4.1.5}$$

Substitution of (4.1.5) in (4.1.2a) yields

$$\begin{aligned} \tilde{x}(t) &= \sum_{k=-\infty}^{\infty} \frac{1}{T_0} X(k\omega_0) e^{jk\omega_0 t} \\ &= \frac{1}{2\pi} \sum_{k=-\infty}^{\infty} X(k\omega_0) e^{jk\omega_0 t}\, \omega_0 \end{aligned} \tag{4.1.6}$$

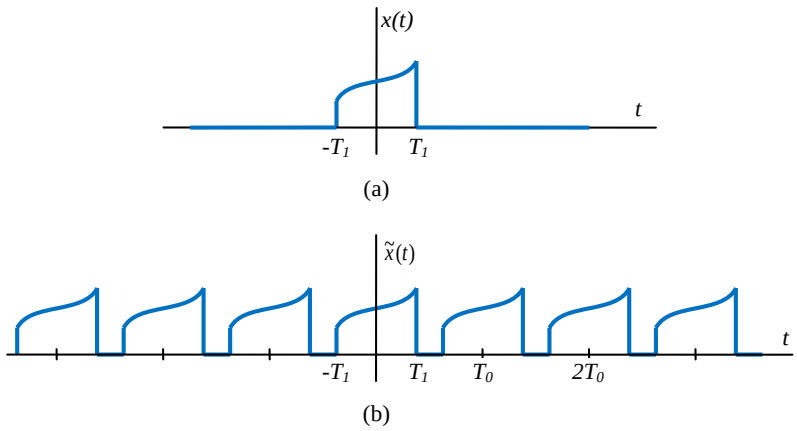

Fig. 4.1.3. (a) Aperiodic signal $x(t)$, (b) periodic signal $\tilde{x}(t)$ which is equal to $x(t)$ over one period.

since $2\pi/T_0 = \omega_0$. In the limit as T_0 approaches infinity, ω_0 becomes $d\omega$, $k\omega_0$ becomes ω and the infinite sum becomes an integral. Recalling that at the limit $\tilde{x}(t) = x(t)$, (4.1.6) becomes

$$x(t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} X(\omega)e^{j\omega t}\, d\omega \tag{4.1.7}$$

where

$$X(\omega) = \int_{-\infty}^{\infty} x(t)e^{-j\omega t}dt \quad . \tag{4.1.8}$$

$X(\omega)$, defined by (4.1.8) is called the Fourier transform or the Fourier integral of the aperiodic signal $x(t)$ and (4.1.7) is the equation for the inverse Fourier transform. The role of the synthesis equation (4.1.7) is similar to that of (3.2.5), in the sense that both express how the signal can be reconstructed from complex exponentials. For periodic signals they have amplitudes a_k and a discrete set of harmonically related frequencies ω_0 . For aperiodic signals however, the complex exponentials occur at a continuum of frequencies and accordingly, the spectrum $X(\omega)$ defined by (4.1.8) becomes a continuous function of ω, as opposed to the discrete nature of a_k vs. k.

4.2. CONVERGENCE OF FOURIER TRANSFORMS

In section 4.1 we have assumed that $x(t)$ was of finite duration. However a broad class of signals of infinite duration also have Fourier transforms. In fact a set of conditions like those required for Fourier series also apply here.

What one would require is that (4.1.7) converges to $x(t)$, when $X(\omega)$ is as given by (4.1.8)

Like the case for periodic signals, there is a set of conditions, again referred to as the Dirichlet conditions, to guarantee that (4.1.7) is equal to $x(t)$ at all points of continuity, and to the average value at points of discontinuity:

i) $x(t)$ is absolutely integrable, i.e.

$$\int_{-\infty}^{\infty} |x(t)|dt < \infty \ . \tag{4.2.1}$$

ii) $x(t)$ has a finite number of maxima and minima within any finite interval .

iii) $x(t)$ has a finite number of discontinuities within any finite interval. Furthermore each discontinuity is finite.

Dirichlet conditions are only sufficient for the existence and convergence of Fourier transforms. An alternative sufficient condition is that x(t)is square integrable, i.e.

$$\int_{-\infty}^{\infty} |x(t)|^2 dt < \infty \ . \tag{4.2.2}$$

However these conditions are by no means necessary. In the sections to follow, we will handle periodic signals which satisfy neither of the above conditions but still have Fourier transforms provided impulse functions are allowed in the transform.

4.3. EXAMPLES OF FOURIER TRANSFORMS

In this section we will attempt to initiate a table of Fourier transform pairs, which, together with the properties of Fourier transforms to be discussed later, becomes handy for many non-trivial problems

Example 4.3.1.

Lets consider

$$.x(t) = e^{-\alpha t}u(t)$$

The Fourier transform can be computed as

$$X(\omega) = \int_0^{\infty} e^{-\alpha t}e^{-j\omega t}dt = \frac{1}{\alpha + j\omega}$$

if $\alpha > 0$.

Note that $X(\omega)$ *of example 4.3.1. is a complex function of* ω. *To plot it as a function of* ω, *we express* $X(\omega)$ *in terms of its magnitude,*

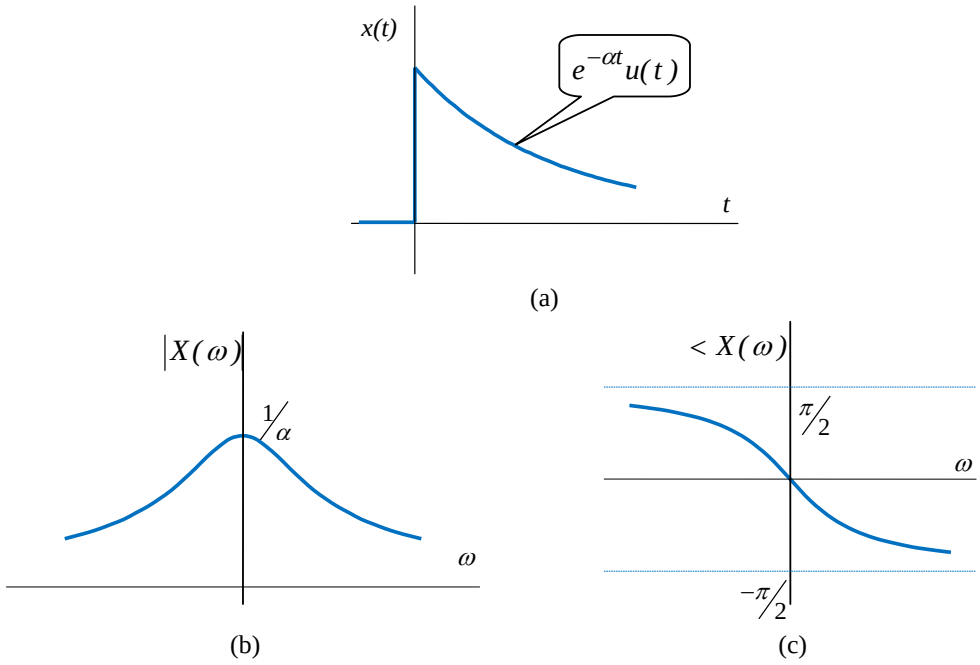

Fig. 4.3.1. Example 4.3.1

$$|X(\omega)| = \frac{1}{\sqrt{\alpha^2 + \omega^2}}$$

and phase

$$< X(\omega) = tan^{-1}\frac{\omega}{\alpha}.$$

These components are sketched in Figure 4.3.1.

Note that when α *is complex,* $x(t)$ *is absolutely integrable if* $\mathcal{R}e\{\alpha\} > 0$ *and again*

$$X(\omega) = \frac{1}{\alpha + j\omega} \quad , \quad \mathcal{R}e\{\alpha\} > 0 \quad .$$

Example 4.3.2.

Let us determine the Fourier transform of the unit impulse

$$x(t) = \delta(t).$$

From 4.1.8

$$X(\omega) = \int_{-\infty}^{\infty} \delta(t)e^{-j\omega t} = 1$$

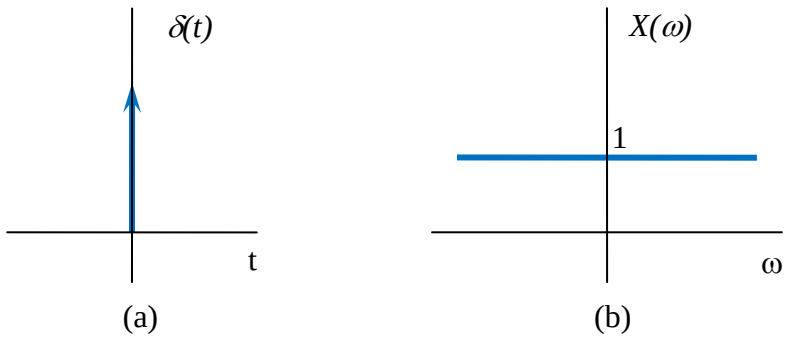

Fig. 4.3.2. Unit impulse and its Fourier transform

Note that, the Fourier transform of the unit impulse consists of equal contributions at all frequencies.

Example 4.3.3.

Let us consider the rectangular pulse

$$x(t) = \begin{cases} 1 & |t| < T \\ 0 & |t| > T \end{cases}$$

as shown in Figure 4.3-3(a). Applying (4.1.7) we have

$$X(\omega) = \int_{-T}^{T} e^{-j\omega t} dt = -\frac{1}{j\omega} e^{-j\omega t}\Big|_{-T}^{T}$$
$$= \frac{2 \sin \omega T}{\omega}$$

which is shown in Figure 4.3.3(b).

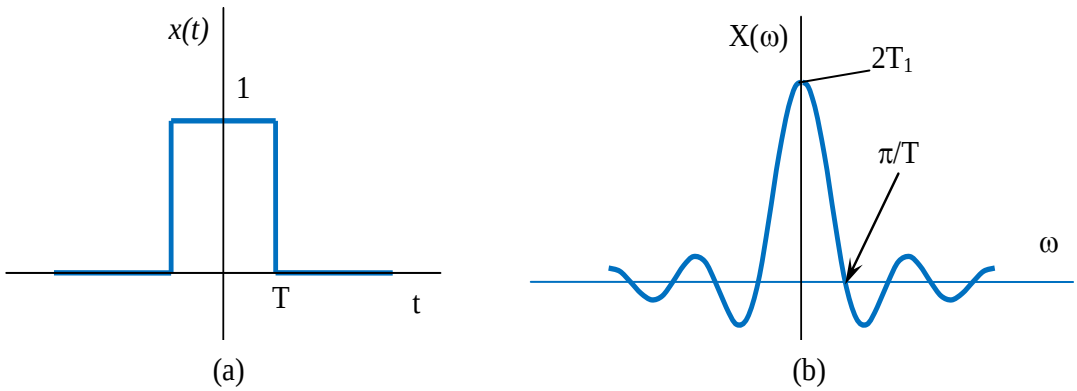

Fig. 4.3.3. The rectangular pulse and its Fourier transform

4.4. FOURIER TRANSFORM OF PERIODIC SIGNALS

So far we have developed the Fourier transform of aperiodic signals by considering the limiting case of Fourier series as the period becomes arbitrarily large. So, there is a strong relationship between the Fourier series and the Fourier transform. In this section we will elaborate more deeply into this relationship and develop the Fourier transform of periodic signals.

Let us recall our important observation about the Fourier series of the rectangular pulse train: the coefficients of the Fourier series were the samples taken from an

envelope which turned out to be the Fourier transform of a single rectangular pulse. This observation can be extended to all periodic signals. Let $\tilde{x}(t)$ be a periodic signal with period T , as shown in Figure 4.4.l(a). Now define $x(t)$ as a signal which is equal to a single period of $\tilde{x}(t)$, i.e.

$$x(t) = \begin{cases} \tilde{x}(t) & -{}^{T_0}/_2 \le t < {}^{T_0}/_2 \\ 0 & otherwise \end{cases} \tag{4.4.1}$$

as depicted in Figure 4.4.l(b).

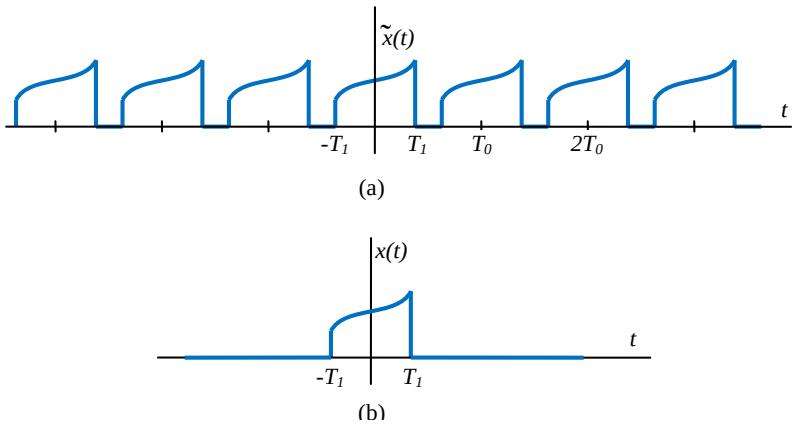

Fig. 4.4.1. A periodic signal and a signal which is equal to its single period.

When one considers the Fourier series coefficients for $\tilde{x}(t)$,

$$\begin{aligned} a_k &= \frac{1}{T_0}\int_{-T_0/2}^{T_0/2} \tilde{x}(t)e^{-jk\omega_0 t}dt = \frac{1}{T_0}\int_{-T_0/2}^{T_0/2} x(t)e^{-jk\omega_0 t}dt \\ &= \frac{1}{T_0}\int_{-\infty}^{\infty} x(t)e^{-jk\omega_0 t}dt = \frac{1}{T_0}X(k\omega_0) \end{aligned} \tag{4.4.2}$$

where $X(\omega)$ is the Fourier transform of $x(t)$. Therefore the Fourier series coefficients of a periodic signal can be obtained from the Fourier transform of a function equal to a single period of the periodic signal by taking samples at intervals of $\omega_0 = 2\pi/T_0$ and dividing by T_0 .

Now we would like to develop Fourier transforms for periodic signals. Such a transform, as we shall see, is possible if impulses are allowed in the transform.

First let us consider a signal $x(t)$, the Fourier transform of which is an impulse of strength 2π at $\omega = \omega_0$, i.e.

$$X(\omega) = 2\pi\delta(\omega - \omega_0). \tag{4.4.3}$$

This signal $x(t)$ can be recovered using the inversion formula as

$$x(t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} 2\pi\delta(\omega - \omega_0)e^{j\omega t}d\omega = e^{j\omega_0 t} \tag{4.4.4}$$

More generally, if

$$X(\omega) = \sum_{k=-\infty}^{\infty} 2\pi.a_k.\delta(\omega - k\omega_0) \tag{4.4.5}$$

then (4.4.5) yields

$$x(t) = \sum_{k=-\infty}^{\infty} a_k \, e^{jk\omega_0 t} \quad , \tag{4.4.6}$$

which is nothing but the Fourier series representation of the periodic signal $x(t)$. Hence, the Fourier transform of a periodic signal is a *weighted impulse train occuring at integer multiples of the fundamental frequency* ω_0, *the weighting factor of the impulse at* $k\omega_0$ *being equal to* 2π *times the k'th Fourier series coefficient* a_k.

Example 4.4.1

Let us consider the sawtooth wave shown in Figure 4.4.2(a). In Example 3.3.1, the Fourier series coefficients for this signal were determined to be

$$a_n = \begin{cases} \dfrac{j}{2\pi n}(-1)^n & n \neq 0 \\ 0 & n = 0 \end{cases}$$

Therefore the Fourier transform of this signal is purely imaginary and consists of impulses occuring at $\omega = k\omega_0$ *with strength* $j(-1)^n/n$. *Figure 4.4.2(b) depicts* $jX(\omega)$.

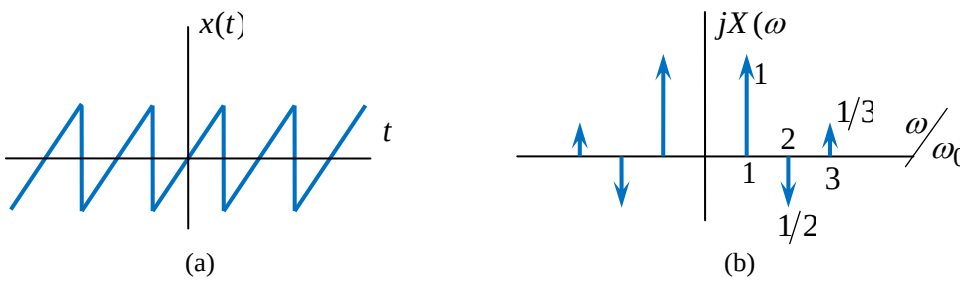

Fig. 4.4.2. Example 4.4.1

Example 4.4.2.

The signal

$$x(t) = \sin \omega_0 t$$

has Fourier series coefficients

$$a_k = \begin{cases} {}^{1}/_{2j} & k = 1 \\ {}^{-1}/_{2j} & k = -1 \\ 0 & \text{otherwise} \end{cases}.$$

Therefore its Fourier transform is

$$X(\omega) = j\pi\{\delta(\omega + \omega_0) - \delta(\omega - \omega_0)\}.$$

Similarly

$$x(t) = \cos \omega_0 t$$

has the transform

$$X(\omega) = \pi\{\delta(\omega + \omega_0) + \delta(\omega - \omega_0)\} \quad .$$

Figure 4.4.3 illustrates these cases.

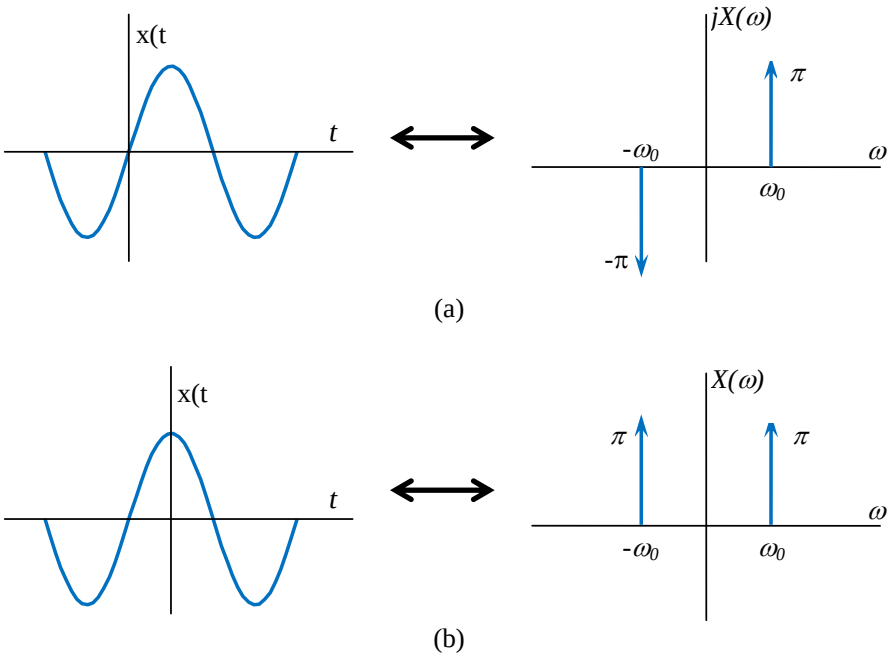

Fig. 4.4.3. Example 4.4.2; Fourier transform of sinusoids

Example 4.4.3

A rather important periodic signal is the periodic impulse train;

$$x(t) = \sum_{k=-\infty}^{\infty} \delta(t - kT)$$

shown in Figure 4.4.4(a). The Fourier series coefficients of this function are

$$a_k = \frac{1}{T}\int_{-T/2}^{T/2} \delta(t) e^{-jk\omega_0 t}$$

Therefore its Fourier transform is a periodic impulse train of impulses of period T and strength $2\pi/T$ as shown in Figure 4.4.4(b).

4.5. PROPERTIES OF FOURIER TRANSFORMS

As mentioned in the preceeding discussion, it is often not necessary to use the defining integral (4.1.8) to evaluate the Fourier transform of a given time

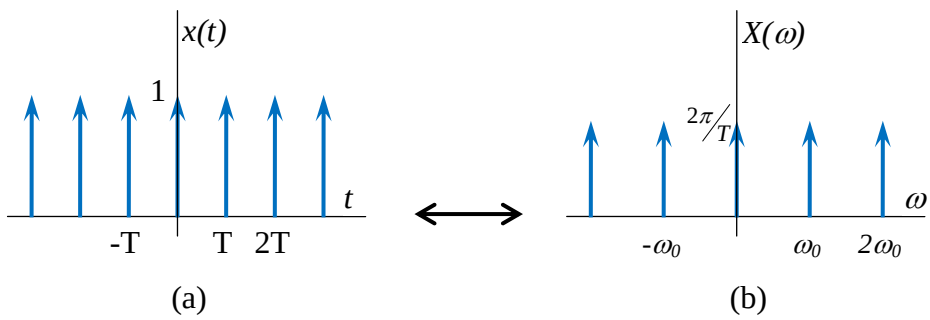

Fig. 4.4.4. Example 4.3.3; Periodic impulse train and its Fourier transform

function. Instead, a table of Fourier transform pairs proves to be more practical.

It is possible to keep this table of transform pairs rather short due to some nice properties of the Fourier transforms. New transform pairs can be generated by using some of these properties on the basic transform pairs, found in the table.

On the other hand, some of these properties make the Fourier transforms useful in the analysis of LTI systems. In this section we will consider a number of the Fourier transforms.

Throughout our discussion, for the sake of convenience we will use a shorthand notation to indicate a relationship between a signal and its transform. We will sometimes refer to the Fourier transform X(ω) of the time signal x(t) with the notation $\mathcal{F}\{x(t)\}$. We will also indicate a Fourier transform pair with the notation

$$x(t) \overset{\mathcal{F}}{\leftrightarrow} X(\omega) \tag{4.5.1}$$

In this last notation, the symbol $\mathcal{F}$ can be omitted if it is clear from the context that the transformation in question is actually the Fourier transform. Thus, for Example 4.3.2;

$$\mathcal{F}\{\delta(t)\} = 1 \ ,$$
$$\mathcal{F}^{-1}\{1\} = \delta(t) \ ,$$

and

$$\delta(t) \overset{\mathcal{F}}{\leftrightarrow} 1 \, .$$

4.5.1. Linearity

It can easily be shown using the defining integral, that

$$\mathcal{F}\{a.x_1(t) + b.x_2(t)\} = a.\mathcal{F}\{x_1(t)\} + b.\mathcal{F}\{x_2(t)\} \tag{4.5.2}$$

or

$$a.x_1(t) + b.x_2(t) \overset{\mathcal{F}}{\leftrightarrow} a.X_1(\omega) + b.X_2(\omega). \tag{4.5.3}$$

It is because of this property that a linear differential equation describing a LTI system transforms into a linear algebraic equation under Fourier transform. (It will later be shown that a differential equation really transforms into an algebraic equation.)

4.5.2. Symmetry Properties

If x(t) is a real valued function, the conjugate of the Fourier transform is

$$\begin{aligned} X^*(\omega) &= \left\{ \int_{-\infty}^{\infty} x(t).e^{-j\omega t} dt \right\}^* \\ &= \int_{-\infty}^{\infty} x(t).e^{j\omega t} dt = X(-\omega) \end{aligned} \tag{4.5.4}$$

As a consequence of this property, the real part of the transform, is an even function of ω whereas the imaginary part is an odd function when $x(t)$ is real. As a matter of fact one can easily show that the real part of $X(\omega)$is the transform of the even part of $x(t)$ and the odd part of $x(t)$ has the transform j times the imaginary part of $X(\omega)$.

As an additional consequence, when $x(t)$ is both real and even, $X(\omega)$ will be real and even. If, on the other hand $x(t)$ is real and odd, $X(\omega)$ will be an odd and purely imaginary function. It is interesting to note the paralelism between these results and symmetry properties of Fourier series.

Example 4.5.1

It was shown earlier (Example 4.3.1) that

$$x(t) = e^{-\alpha t}u(t) \leftrightarrow X(\omega) = \frac{1}{\alpha + j\omega} \quad .$$

Noting that the even part of $x(t)$ is

$$x_e(t) = \frac{1}{2}\{e^{-\alpha t}u(t) + e^{\alpha t}u(-t)\} = \frac{1}{2}e^{-\alpha|t|}$$

and the real part of the transform is

$$\mathcal{R}e\{X(\omega)\} = \frac{\alpha}{\alpha^2 + \omega^2} \quad ,$$

one has a new transform pair

$$e^{-\alpha|t|} \leftrightarrow \frac{\alpha}{\alpha^2 + \omega^2} \quad .$$

4.5.3. Time Shifting

If

$$x(t) \overset{F}{\leftrightarrow} X(\omega) \tag{4.5.5}$$

then

$$x(t - t_0) \leftrightarrow \int_{-\infty}^{\infty} x(t - t_0).e^{-j\omega t}dt = \int_{-\infty}^{\infty} x(\mu).e^{-j\omega(\mu + t_0)}d\mu$$
$$= e^{-j\omega t_0}X(\omega)$$

i.e.

$$x(t - t_0) \leftrightarrow e^{-j\omega t_0}X(\omega) \tag{4.5.6}$$

When the transform $X(\omega)$ is expressed in polar form

$$X(\omega) = |X(\omega)|.e^{j\theta(\omega)} \, , \tag{4.5.7}$$

it is clearly seen that, shifting a signal $x(t)$ in time does not alter the magnitude of its transform but introduces a phase shift since

$$x(t - t_0) \leftrightarrow e^{-j\omega t_0}|X(\omega)|.e^{j\theta(\omega)} = |X(\omega)|.e^{j[\theta(\omega) - \omega t_0]} \tag{4.5.8}$$

4.5.4. Time and Frequency Scaling

When $x(t)$ and $X(\omega)$ is a Fourier transform pair, then

$$x(at) \leftrightarrow \frac{1}{|a|}X\left(\frac{\omega}{a}\right) \tag{4.5.9}$$

where a is a real constant. This follows directly from the defining integral since

$$x(at) \leftrightarrow \int_{-\infty}^{\infty} x(at).e^{-j\omega t}dt = \int_{-\infty}^{\infty} x(\sigma).e^{-j\omega\sigma/a}\, d\sigma/a$$
$$= \begin{cases} \frac{1}{a}X(\frac{\omega}{a}) & a > 0 \\ \frac{-1}{a}X(\frac{\omega}{a}) & a < 0 \end{cases} \quad . \tag{4.5.10}$$

We note from this result that contracting the time scale amounts to expanding the frequency scale and vice versa. This is another example of the inverse relationship between time and frequency domains. A signal that varies rapidly within a small time interval will have a transform that occupies a wider frequency band. The output of a system which does not pass high frequency components, will have relatively slow changes compared to the input.

4.5.5. Differentiation

Let x(t) and X(ω) be a Fourier transform pair. By differentiating both sides of the synthesis equation (4.17) one obtains

$$\frac{dx(t)}{dt} = \frac{1}{2\pi}\int_{-\infty}^{\infty} j\omega X(\omega)e^{j\omega t}d\omega \tag{4.5.11}$$

which implies that

$$\frac{dx(t)}{dt} \leftrightarrow j\omega X(\omega) \ . \tag{4.5.12}$$

This is a very important property, which makes the Fourier transforms so valuable in the solution of linear differential equations: differentiation in the time domain corresponds to multiplication by jω in the frequency domain.

Transforming both sides of the general linear, time invariant differential equation

$$\sum_{k=0}^{N} a_k \frac{d^k y(t)}{dt^k} = \sum_{k=0}^{M} b_k \frac{d^k x(t)}{dt^k} \tag{4.5.13}$$

yields

$$\left\{\sum_{k=0}^{N} a_k (j\omega)^k\right\} Y(\omega) = \left\{\sum_{k=0}^{N} b_k (j\omega)^k\right\} X(\omega) \tag{4.5.14}$$

where $Y(\omega)$ and $X(\omega)$ are the Fourier transforms of the output $y(t)$ and the input $x(t)$ respectively. (4.5.14) can be solved for $Y(\omega)$, the unknown, in terms of the input $X(\omega)$. Thus, the remaining task is to determine the inverse transform of $Y(\omega)$ to find the solution $y(t)$ in the time domain.

Example 4.5.2

Let us determine the impulse response of the LTI system described by the differential equation

$$\frac{dy(t)}{dt} + \alpha y(t) = x(t)$$

Fourier transform of both sides of this equation yields

$$(j\omega + \alpha)Y(\omega) = X(\omega)\,,$$

which can be solved for $Y(\omega)$ *as*

$$Y(\omega) = \frac{X(\omega)}{j\omega + \alpha} \ .$$

Since

$$x(t) = \delta(t) \leftrightarrow X(\omega) = 1\,,$$

the transform of the impulse response is

$$H(\omega) = \frac{1}{j\omega + \alpha} \ .$$

Remembering (Example 4.5.1) that

$$e^{-\alpha t}u(t) \leftrightarrow \frac{1}{j\omega + \alpha},$$

we conclude

$$h(t) = e^{-\alpha t}u(t) \ .$$

4.5.6. Integration

Since differentiation in time domain corresponds to multiplication by jω in the frequency domain, one would expect that integration goes together with a division by jω in the frequency domain. But unfortunately this is not the whole picture.

Let

$$g(t) = \int_{-\infty}^{t} x(\tau)d\tau \ . \tag{4.5.15}$$

For the above expectation to be true, i.e., for

$$G(\omega) = \frac{1}{j\omega}X(\omega) \tag{4.5.16}$$

to hold, the transform must exist in the first place. However G(ω) given by (4.5.16) blows up at the origin unless X(0) = 0. Thus (4.5.16) is valid only if

$$X(0) = \int_{-\infty}^{\infty} x(\tau)d\tau = 0 \ . \tag{4.5.17}$$

In the more general case where (4.5.17) does not hold, the precise relationship is

$$\mathrm{G}(\omega) = \frac{1}{\mathrm{j}\omega}\mathrm{X}(\omega) + \pi\mathrm{X}(0)\delta(\omega) \ , \tag{4.5.18}$$

the impulse term on the right hand side accounting for the dc component that results from the integration.

To understand this property better, let us consider the unit step signal $u(t)$. This signal can be expressed as the sum of its even part $x_e(t) = 1/2$ and its odd part $x_o(t) = u(t) - 1/2$. The odd part satisfies (4.5.17) and

$$\frac{d}{dt}x_o(t) = \delta(t) \leftrightarrow 1 \ . \tag{4.5.19}$$

Thus

$$x_o(t) \leftrightarrow \frac{1}{j\omega} \ . \tag{4.5.20}$$

If we consider (4.4.3) and (4.4.4)

$$x_e(t) = 1/2 \leftrightarrow \pi.\delta(\omega). \tag{4.5.21}$$

Thus we obtain the important transform pair

$$u(t) \leftrightarrow \frac{1}{j\omega} + \pi.\delta(\omega) \tag{4.5.22}$$

4.5.7. Duality

Let us repeat the analysis and synthesis equations here once again to observe this nice property:

$$X(\omega) = \int_{-\infty}^{\infty} x(t)e^{-j\omega t}dt \ , \tag{4.5.23a}$$

$$x(t) = \int_{-\infty}^{\infty} X(\omega)e^{j\omega t}d\omega \ . \tag{4.5.23b}$$

A close inspection shows that there is a definite symmetry in these equations. If we interchange ω and t we get almost the same equations except for a factor of 2π and some sign changes.

Earlier we have used the two transform pairs

$$\delta(t) \leftrightarrow 1 \tag{4.5.24a}$$

and

$$1 \leftrightarrow 2\pi.\delta(t) \ . \tag{4.5.24b}$$

These transform pairs could actually be obtained from the duality property.

To put the duality property on a more sound basis,. let us consider two functions related by the integral relationship

$$f(u) = \int_{-\infty}^{\infty} g(v)e^{-juv}dv \ . \tag{4.5.25}$$

When (4.5.25) is compared with (4.5.23), it is observed that with $u = \omega$ and $v = t$,

$$f(\omega) = \mathcal{F}\{g(t)\} \tag{4.5.26}$$

whereas with $u = t$ and $v = \omega$,

$$g(-\omega) = \frac{1}{2\pi}\mathcal{F}\{f(t)\} \ . \tag{4.5.27}$$

Thus if we are given the Fourier transform pair

$$g(t) \leftrightarrow f(\omega) \tag{4.5.28}$$

then the time function f(t) has the transform

$$f(t) \leftrightarrow 2\pi.g(-\omega) \tag{4.5.29}$$

Example 4.5.3

Since

$$x(t) = \delta(t) \leftrightarrow X(\omega) = 1$$

from duality one obtains

$$X(t) = 1 \leftrightarrow 2\pi.x(-\omega) = 2\pi.\delta(\omega)$$

verifying (4.4.3).

Example 4.5.4

Suppose that we want to determine the Fourier transform of

$$x(t) = \frac{1}{1+t^2} \ .$$

Recalling (Example 4.5.1) that

$$e^{\alpha|t|} \leftrightarrow \frac{2\alpha}{\alpha^2+\omega^2} \ ,$$

by duality

$$\frac{2\alpha}{\alpha^2+t^2} \leftrightarrow 2\pi.e^{-\alpha|\omega|} \quad .$$

Therefore letting α=1, and multiplying both sides by 1/2

$$\frac{1}{1+t^2} \leftrightarrow \pi.e^{-|t|} \, .$$

The duality property can be used to derive some aditional properties. To be more specific, duals of the properties presented so far can be obtained easily. Proofs of these properties listed below are left to the reader as an excercise.

Frequency shifting:

$$e^{j\omega_0 t}.x(t) \leftrightarrow X(\omega-\omega_0) \, , \tag{4.5.30}$$

Frequency differentiation:

$$-jtx(t) \leftrightarrow \frac{d}{d\omega}X(\omega) \, , \tag{4.5.31}$$

Frequency integration:

$$-\frac{1}{jt}x(t)+\pi.x(0).\delta(t) \leftrightarrow \int_{-\infty}^{\omega} X(\mu)d(\mu) \, . \tag{4.5.32}$$

4.5.8. The Convolution Property

One of the most important properties of the Fourier transform is about the convolution operation for its important application in the analysis of LTI systems.

Let $x(t)$ be the input to a LTI system whose impulse response is h(t). Since x(t) can be expressed as

$$x(t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} X(\omega)e^{j\omega t} \tag{4.5.33}$$

in terms of its Fourier transform, the output is

$$\begin{aligned} y(t) = x(t) * h(t) &= \int_{-\infty}^{\infty} h(\tau)\left[\frac{1}{2\pi}\int_{-\infty}^{\infty} X(\omega)e^{j\omega(t-\tau)}\right]d\tau \\ &= \frac{1}{2\pi}\int_{-\infty}^{\infty}\left[\int_{-\infty}^{\infty} h(\tau).e^{-j\omega\tau}d\tau\right]X(\omega).e^{j\omega t}d\omega \end{aligned} \tag{4.5.34}$$

The integral inside the square brackets is nothing but the Fourier transform $H(\omega)$ of the impulse response $h(\tau)$. Therefore

$$y(t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} H(\omega)X(\omega)e^{j\omega t}d\omega \quad . \tag{4.5.35}$$

Since y(τ) and its transform $Y(\omega)$ are related by

$$y(t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} Y(\omega)e^{j\omega t} \tag{4.5.36}$$

Y(ω) can be identified as

$$Y(\omega) = X(\omega)H(\omega) \, . \tag{4.5.37}$$

That is

$$x(t) * h(t) \leftrightarrow X(\omega)H(\omega) \tag{4.5.38}$$

This property is essentially a consequence of the fact that complex exponentials are eigen functions of LTI systems. $H(\omega)$, the eigenvalue associated with the eigenfunction $e^{j\omega t}$, or the Fourier transform of the impulse response is simply the change in complex amplitude of the complex exponential as it passes through the LTI system. $H(\omega)$ is generally referred to as the frequency response, the transfer function or the system function of the system and it plays as important a role as its inverse transform $h(t)$. Since $h(t)$ characterizes an input-output relationship fully, so does its transform $H(\omega)$. Many of the properties of LTI systems can be explained conveniently interms of $H(\omega)$. For example, it is now a trivial problem to show that a cascade connection of single input/single output LTI systems has the commutative property, since multiplication is a commutative operation.

It is important that the system function $H(\omega)$ is not defined for every LTI system. For $H(\omega)$ to exist, $h(t)$ must have a Fourier transform. If $h(t)$ satisfies the three Dirichlet conditions for example, the first condition will require $h(t)$ to be absolutely integrable, which is equivalent to saying that the system is stable. However, note that Dirichlet conditions are only sufficient conditions, and in the case the system is unstable, the Fourier transform just may not exist.

Thus, in using Fourier transforms for the analysis of LTI systems, we will consider systems with impulse responses that possess Fourier transforms only. For the investigation of LTI systems that do not have finite-valued transfer functions we will have to generalize Fourier transforms to Laplace transforms as we are going to do in the chapters to follow.

Example 4.5.5

Consider the LTI system with

$$h(t) = \delta(t - t_0)\,.$$

The frequency response of this system is

$$H(\omega) = \mathcal{F}\{\delta(t - t_0)\} = e^{-j\omega t_0}.$$

Therefore

$$Y(\omega) = X(\omega)H(\omega) = e^{-j\omega t_0}X(\omega).$$

Note that this result is consistent with the fact that this system is a pure delay since

$$x(t - t_0) \leftrightarrow e^{-j\omega t_0}X(\omega)$$

according to (4.5.6).

Example 4.5.6

As a second example, let us consider a capacitor as a system and define the input as the voltage across and the output as the current through. In this case

$$x(t) = C\frac{dx(t)}{dt}\,.$$

From the differentiation property (4.5.12)

$$Y(\omega) = Cj\omega X(\omega)$$

or

$$H(\omega) = \frac{Y(\omega)}{X(\omega)} = j\omega C\,.$$

Recalling that $Y(\omega)$ stands for current through and $X(\omega)$ for the voltage accross, $H(\omega)$ is the ratio of transform of the current to that of the voltage. In fact that is what we know as the admittance of a capacitor.

Example 4.5.7

Let us consider the circuit of Figure 4.5.1(a).

The loop equation for this circuit yields

$$\frac{dy(t)}{dt} + y(t) = x(t) \quad .$$

Transforming both sides, one obtains the algebraic equation

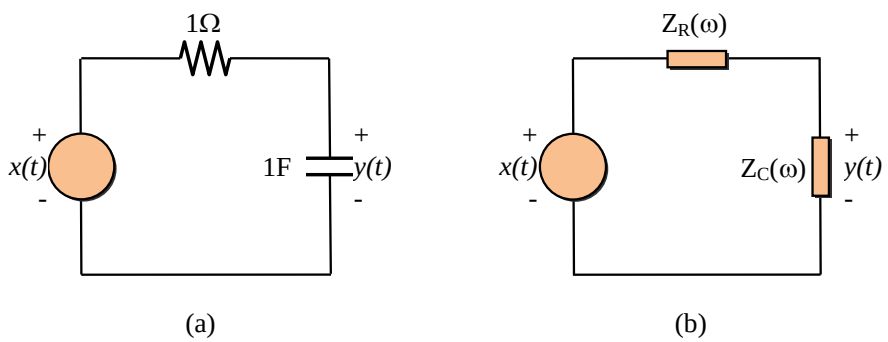

Fig. 4.5.1. Circuit for Example 4.5.7. (a) in the time domain, (b) in the frequency domain

$$(j\omega + 1)Y(\omega) = X(\omega)$$

and thus the frequency response is

$$H(\omega) = \frac{Y(\omega)}{X(\omega)} = \frac{1}{j\omega + 1} \quad .$$

This result could be obtained more easily, if the passage to the frequency domain was performed at the beginning. The frequency domain equivalent of the circuit is shown in Figure 4.5.1(b). Here $Z_R(\omega)$ represents the "impedance" of the resistor defined as the ratio of the transform of the voltage across to that of the current through. Hence $Z_R(\omega) = R = 1$. $Z_C(\omega)$ is defined for the capacitor in the same way and (See Example 4.5.5) $Z_C(\omega) = {}^{1}/_{j\omega C}$. Thus solving for $Y(\omega)$ by simple potential division, one obtains the same frequency response.

Example 4.5.8.

Let us determine the response of the system of Example 4.5.7 to the signal of Example 3.2.1., rewritten here for convenience:

$$x(t) = \sum_{k=-3}^{3} a_k . e^{jkt}$$

with a0=1, a1=a-1=3/4, a2=a-2=-l/2 and a3=a-3=l/4.

Then,

$$X(\omega) = \sum_{k=-3}^{3} 2\pi . a_k . \delta(\omega - k)$$

and since

$$H(\omega) = \frac{1}{j\omega + 1} \quad ,$$

$$Y(\omega) = X(\omega)H(\omega) = \sum_{k=-3}^{3} 2\pi . a_k \frac{1}{1+j\omega}\delta(\omega - k)$$
$$= \sum_{k=-3}^{3} 2\pi . a_k \frac{1}{1+jk}\delta(\omega - k) \quad .$$

Converting this back to the time domain, we obtain

$$y(t) = \sum_{k=-3}^{3} \frac{a_k}{1+jk} . e^{jkt} \quad ,$$

which is the Fourier series for the output and is identical with the result of Example 3.2.2.

4.5.9. The Modulation Property

According to the convolution property, convolution of two signals in the time domain corresponds to multiplication in the frequency domain. From the duality between the time and frequency domains, multiplication in the time domain corresponds to convolution in the frequency domain. Specifically

$$f(t).g(t) \longleftrightarrow \frac{1}{2\pi} F(\omega) * G(\omega) \, . \tag{4.5.39}$$

Multiplication of one signal by another can be interpreted as using one signal to scale or modulate the amplitude of the other. Thus such a multiplication is often referred to as amplitude modulation.

Example 4.5.9

Let $f(t)$ be a signal, with a spectrum $F(\omega)$ as shown in Figure 4.5.2(a) .Also consider the signal $g(t)$ defined by

$$g(t) = cos\, \omega_0 t \, .$$

Then,

$$G(\omega) = \pi . \delta(\omega - \omega_0) + \pi . \delta(\omega + \omega_0)$$

as depicted in Figure 4.5.2(b) .

The spectrum of the product is obtained by an application of (4.5.39) yielding

$$s(t) = f(t).g(t) \longleftrightarrow S(\omega) = \frac{1}{2\pi} F(\omega) * G(\omega)$$
$$= \tfrac{1}{2} F(\omega - \omega_0) + \tfrac{1}{2} F(\omega + \omega_0)$$

The signal $s(t) = f(t).g(t)$ and its spectrum are shown in Figure 4.5.2(c). Here we see that the spectrum $S(\omega)$ of the product (the modulated signal) consists of the sum of two shifted and scaled versions of $F(\omega)$. Note that, during the modulation, all the information in $f(t)$ is preserved. Only, this information is shifted to higher frequencies, which is more suitable for transmission.

4.5.10. Parsevals Formula

Often in applications, the total energy of a signal $x(t)$, defined as the integral over all times of $|x(t)|^2$ is required. This can be done in a straight forward manner by determining the energy per unit time ($|x(t)|^2$) and integrating from minus infinity to plus infinity. On the other hand, the frequency domain representation of $x(t)$ contains all the information $x(t)$ carries, including its energy. In other words the total energy in $x(t)$ can be calculated by determining the energy per unit frequency ($|X(\omega)|^2/2\pi$) and integrating over all frequencies.

To prove this relation let us rewrite (4.5.39) as

$$\int_{-\infty}^{\infty} f(t)g(t)e^{-j\omega t}dt = \frac{1}{2\pi}\int_{-\infty}^{\infty} F(\mu)G(\omega-\mu)d\mu \qquad (4.5.40)$$

Setting $\omega = 0$, one obtains

$$\int_{-\infty}^{\infty} f(t)g(t)dt = \frac{1}{2\pi}\int_{-\infty}^{\infty} F(\mu)G(-\mu)d\mu \qquad (4.5.41)$$

If, in this last equation $g(t) = f^*(t)$, since

$$f^*(t) \leftrightarrow F^*(-\omega) \ , \qquad (4.5.42)$$

with $\mu = \omega$, (4.5.41) becomes

$$\int_{-\infty}^{\infty} f(t)f^*(t)dt = \frac{1}{2\pi}\int_{-\infty}^{\infty} F(\omega)F^*(\omega)d\omega \qquad (4.5.43)$$

or for a signal $x(t)$,

$$\int_{-\infty}^{\infty} |x(t)|^2 dt = \frac{1}{2\pi}\int_{-\infty}^{\infty} |X(\omega)|^2 d\omega \ .$$

This last relation is known as the Parseval's formula.

The function $|X(\omega)|^2$ is often referred to as the energy-density spectrum for it represents the energy distribution over the frequency scale.

The energy in a periodic signal is infinite and therefore (4.5.44) is not valid for such a function. However in such a case, energy in one period of the time function can be related to its spectrum as

$$\frac{1}{T_0}\int_0^{T_0} |x(t)|^2 dt = \sum_{k=-\infty}^{\infty} |a_k|^2 \quad . \qquad (4.5.45)$$

Thus for a periodic signal, Parseval's relation relates the energy in one period of the time function to the energy in Fourier series coefficients. The quantity $|a_k|^2$ is that part of the energy per period, contributed by the k'th harmonic.

Example 4.5.9

Let us compare the input and output powers for Example 3.2.2.

The input power (total energy per period)is

$$\frac{1}{T_0}\int_0^{T_0} |x(t)|^2 dt = \sum_{k=-3}^{3} |a_k|^2$$

$$= \left(\frac{1}{4}\right)^2 + \left(\frac{1}{2}\right)^2 + \left(\frac{3}{4}\right)^2 + 1 + \left(\frac{3}{4}\right)^2 + \left(\frac{1}{2}\right)^2 + \left(\frac{1}{4}\right)^2 = \frac{11}{4}$$

On the other hand the output power is

$$\frac{1}{T_0}\int_0^{T_0} |y(t)|^2 dt = \sum_{k=-3}^{3} |b_k|^2$$

$$= \frac{1}{10}\left(\frac{1}{4}\right)^2 + \frac{1}{5}\left(\frac{1}{2}\right)^2 + \frac{1}{2}\left(\frac{3}{4}\right)^2 + 1 + \frac{1}{2}\left(\frac{3}{4}\right)^2 + \frac{1}{5}\left(\frac{1}{2}\right)^2 + \frac{1}{10}\left(\frac{1}{4}\right)^2 = \frac{17}{10}$$

Hence the "power gain" is

$$\frac{17}{10}\Big/\frac{11}{4} = \frac{34}{55}$$

which accounts for a loss in power, as expected, since the system is a passive circuit with dissipative elements.

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(*) of relatively higher level

PROBLEMS

Problem 1.

A time-invariant system is characterized by the input/output relationship

$$y(t) = \begin{cases} x(t) & x(t) \geq 0 \\ \frac{1}{2}x(t) & x(t) < 0 \end{cases}$$

Determine the Fourier transform of the output $y(t)$ when the input is $x(t) = \sin \omega_0 t$.

Problem 2.

For a real function $f(t)$, the auto-correlation function is defined as

$$g(t) = \int_{-\infty}^{\infty} f(\tau) f(t+\tau) d\tau \quad .$$

Determine the Fourier transform $G(\omega)$ of $g(t)$, in terms of the Fourier transform $F(\omega)$ of $f(t)$, in its simplest form.

Problem 3.

Determine the time function $f(t)$, the Fourier transform of which is

$$F(\omega) = \omega . e^{-\omega^2} \quad .$$

Problem 4.

Using a table of transforms, determine the Fourier transform of the function $f(t)$ shown in Figure P4.

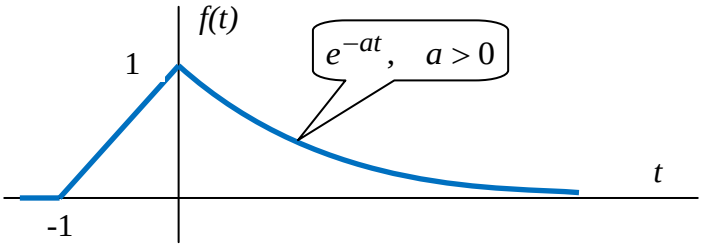

Figure P4

Problem 5.

Determine and sketch the Fourier Transform of the signal considered in Problem 3 of Chapter 3.

Problem 6.

The Fourier transform of the pulse $p(t)$ shown Figure P6 is

$$P(\omega) = \frac{2\,sin\,\omega T_1}{\omega} \quad .$$

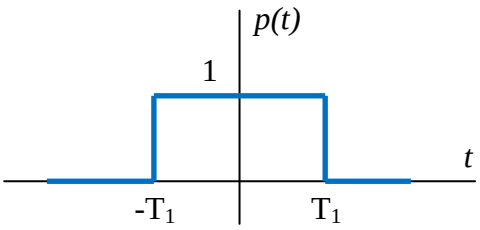

Figure P6

Determine the time function $g(t)$ which has the Fourier transform

$$G(\omega) = u(\omega) - u(\omega - 1) = \begin{cases} 1 & 0 < \omega \leq 1 \\ 0 & \text{otherwise} \end{cases} \quad .$$

Problem 7.

Find the signal $x(t)$, the Fourier transform $X(\omega)$ of which is defined by

$$|X(\omega)| = \begin{cases} e^{-\omega} & 0 \leq \omega \\ 0 & \omega < 0 \end{cases} \quad , \quad \angle X(\omega) = -2\omega \quad .$$

Problem 8

A time function $y(t)$ is defined as $y(t) = x(t) + x(-t + t_0)$, where $x(t)$ is an arbitrary waveform which is nonzero for $0 \leq t \leq T$ and zero elsewhere. If $y(t)$ has a Fourier transform $Y(\omega)$, find its phase $\angle Y(\omega)$.

Problem 9

The response of a LTI system to an input

$$x(t) = e^{-t}u(t)$$

is

$$y(t) = (2e^{-t} - 2e^{-2t})u(t)$$

Find the frequency response $H(\omega)$ and the unit impulse response $h(t)$ of this system. Also find the differential equation describing the system, using the frequency response.

Problem 10

Consider the system shown in Figure P10 where the box labeled 'delay' introduces an analog delay of T seconds.

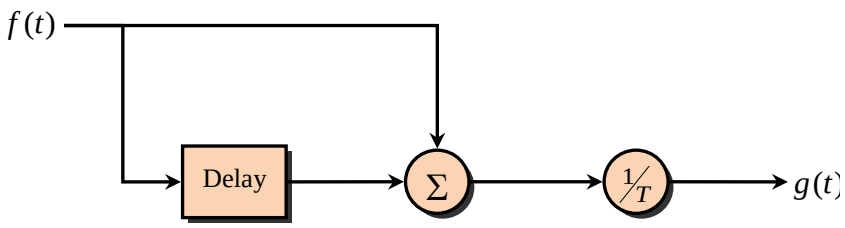

Figure P10

i)Find the Fourier Transform of the outp $g(t)$ut in terms of that of the input $f(t)$.

ii)What operation in time domain does this system perform on its input $f(t)$ when T approaches zero?

Problem 11

A special kind of transform on $x(t)$ is defined as

$$X_S(\omega) = \int_0^{\infty} x(t).\,sin(\omega t).\,dt$$

Determine the relationship between $X_S(\omega)$ and the Fourier transform $X(\omega)$of $x(t)$, for $x(t)$ odd.

Problem 12

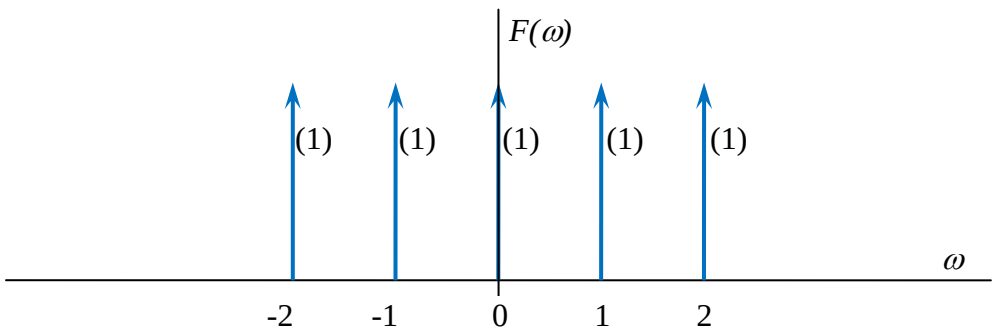

Figure P12

The Fourier transform $F(\omega)$ of a function $f(t)$ is as given in Figure P12. Can you determine the Fourier transform of $f^2(t)$?

Problem 13

Two LTI systems characterized by impulse responses $h_1(t)$ and $h_2(t)$ are band limited to $\omega_1$and ω_2 respectively, (i.e. $H_i(\omega) = 0$ for $|\omega| > \omega_i$).

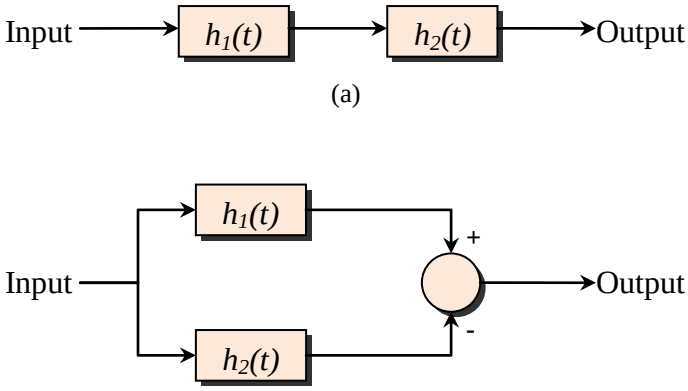

Figure P13

Determine if the impulse responses of the combinations shown in Figure P13 are band limited and if so, to what frequency.